VesselVision: Fleet Safety Awareness over Streaming Vessel Trajectories (Demo Paper)

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ABSTRACT

The massive-scale data generation of positioning (tracking) messages, collected by various surveillance means, has posed new challenges in the field of mobility data analytics in terms of extracting valuable knowledge out of this data. One of these challenges is online maritime awareness, where the goal is to monitor and ensure the safety of a fleet, including, among others, collision risk assessment. To address this challenge, we present VesselVision, a system that estimates, tracks, and visualizes vessels' collision risk. In particular, our system offers a unified solution that tracks vessels that are detected to be in encountering process and assess their corresponding collision risk over streaming AIS position data in an online fashion. The functionality of our system is demonstrated over popular real-world AIS datasets.

CCS CONCEPTS

• **Information systems** → **Data mining**; **Spatial-temporal systems**.

KEYWORDS

Data Analytics, Maritime Safety, Collision Risk Assessment, Machine Learning, Visual Analytics

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1 INTRODUCTION

Mobility Data Analytics (MDA) is a hot interdisciplinary research field, where the ultimate goal is to discover knowledge from spatio-temporal, geo-referenced information [6, 1]. Extracting valuable knowledge out of the sheer amount of data generated by various surveillance means can reveal motion behaviours or patterns that can be exploited in several fields. In this demo paper, we focus on the maritime domain, where huge amounts of tracking data are being generated on a daily basis by vessels sailing worldwide using the ubiquitous Automatic Identification System (AIS). These particular data streams, which are stored for analytics purposes, provide interesting challenges and precious opportunities on critical use-case scenarios.

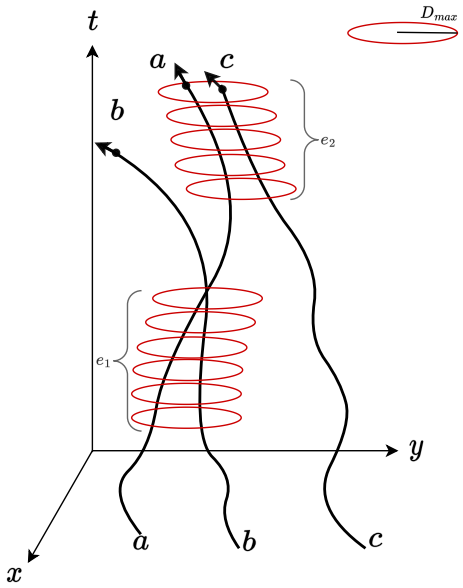


Figure 1: VCRA example for three moving objects a, b and c. Solid lines correspond to vessels' trajectories, whereas red ellipses correspond to the timeslices which the encountering processes e_1 and e_2 were tracked and assessed using the VCRA model.

Accurate Vessel Collision Risk Assessment (VCRA) is one such application, the goal of which is to effectively assess the collision risk in a fleet of monitored vessels. The expected widespread use of unmanned surface vessels (USV) in the near future [2] (as an alternative transportation mode in order to reduce the use of road and rail transportation) poses a plethora of maritime safety concerns. Figure 1 illustrates an example, where we depict the vessels' trajectories (solid black line) and two detected encounterings, namely e_1 (a, b) and e_2 (a, c).

In this context, we demonstrate VesselVision, a comprehensive MDA framework that integrates state-of-the-art analytical techniques and algorithms for collision risk assessment into a unified solution that enhances maritime situational awareness requirements.

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The core of our approach consists of a machine learning-based VCRA model, originally proposed in [8], which is used to calculate the collision risk index of a pair of vessels that are detected to be in an encountering process. The framework is built upon ST_Visions [9], a Python-based interactive Visual Analytics (VA) tool. The functionality of our system is demonstrated through three different use-case scenarios using three real-world datasets, of which two open (Brest [7] and Piraeus [11]) and one closed (MarineTraffic).

2 SYSTEM OVERVIEW

In this section, we describe the modules of our system and how they communicate with each other. Figure 2 illustrates the architecture of VesselVision. We distinguish three layers: the *Data Harvesting and Processing* layer, which includes the online preparation (cleansing and storage) and exploitation of the raw AIS data stream, the *Model Training* layer, which trains the VCRA model [8] based on the historical data of AIS-equipped vessels, and the *Data Visualisation* layer, which visualises the encountering vessels with their associated CRI, based on the ST_Visions VA tool [9]. The aforementioned layer is described in more detail in Section 3.

Data Harvesting and Processing: Our approach initially considers a data pre-processing phase, which consists of two steps: (i) drop duplicate (i.e., records transmitted simultaneously) and erroneous records, based on an upper threshold for vessels' speed (set to 50 knots), and (ii) temporally align the vessels' locations using a temporal buffer for each object in order to compensate for the irregular sampling rate of AIS [1]. In a nutshell, the buffer serves as a temporal window where vessels' locations are temporarily stored in order to be extrapolated at a specific time instance. This is known as temporal alignment, and it ensures that all vessel locations within the buffer are correctly synchronised in terms of time.

Model Training: The current state-of-the-art in VCRA consists of formulaic- as well as Deep Learning (DL) -based approaches. The difference between the two method families is that the former employs kinematic equations in conjunction with ML models (e.g. SVM), whereas the latter employs DL methods (e.g. Convolutional Neural Networks – CNN) to assess the collision risk between two vessels. In order to address the VCRA task, we use a variant of the MLP-VCRA model proposed in [8], in which we use the azimuth angle (α_O^T) and relative direction (ϕ_O^T) of the own with respect to the target vessel, in addition to their corresponding speed (V_O, V_T), course (ϕ_O, ϕ_T), and haversine distance (D_O^T). Compared to previous formulaic-based approaches, our model outperforms them by a wide margin in terms of both quality and latency, as it is experimentally shown in [8]. Given the aforementioned features, the goal of MLP-VCRA is to compute the encountering vessels' Collision Risk Index (CRI), which is a number between zero and one expressing their collision likelihood. In more detail, Eq. 1 illustrates the mathematical formula for calculating the CRI[1], where W denotes the weight vector of target factors, and U denotes the membership vector of target factors, namely the Distance and Time to Closest Point of Approach (DCPA and TCPA, respectively), distance D_O^T, relative bearing θ_O^T, and speed ratio $K = \frac{V_T}{V_O}$ [5].

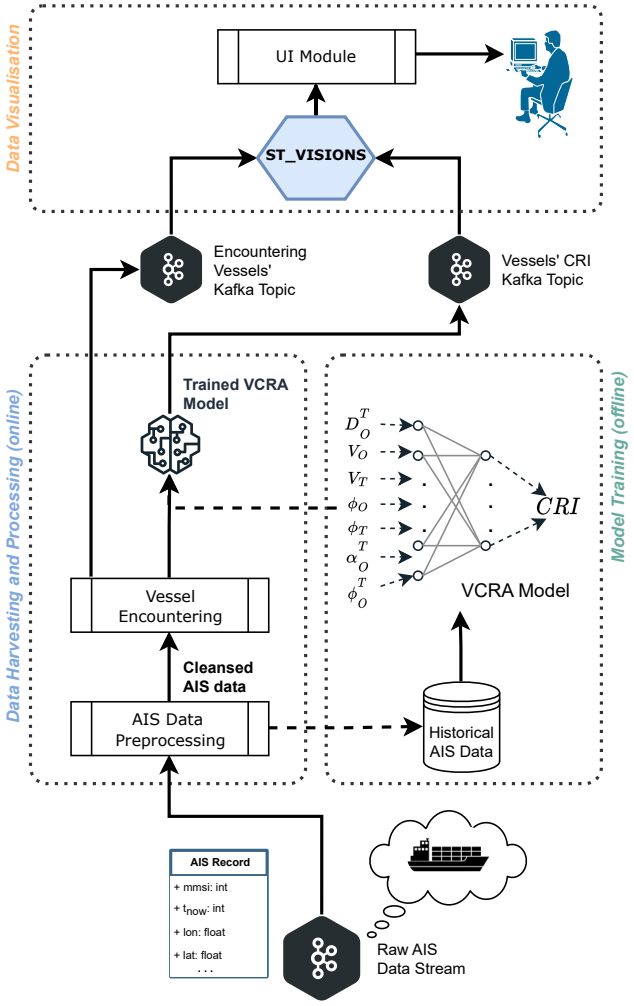


Figure 2: Overview of VesselVision

$$CRI = W \cdot U = W_{DCPA}U_{DCPA} + W_{TCPA}U_{TCPA} + W_{D_O^T}U_{D_O^T} + W_{\theta_O^T}U_{\theta_O^T} + W_K U_K \quad (1)$$

Vessel Encountering: The *Vessel Encountering* method (VE) aims at identifying vessel pairs over (streaming) surveillance data. Two vessels, called own (O) and target (T), respectively, are in an encountering process, when their distance D_O^T is lower than a threshold d_{max} (e.g., 1 n.m.) for a time period t longer than a threshold t_{min} (e.g., 5 min.), with D_O^T monotonically decreasing over the aforementioned period. After the necessary pre-processing (cleansing, etc.) of the raw AIS data stream, and in order to extract interpretable trajectories, we focus on moving own vessels, i.e., those with instant speed V_O higher than a threshold v_{min} (e.g., 1 knot, to address GPS inaccuracy), while target vessels can be moving or stationary.

[1]More information on the mathematical formulae can be found at https://github.com/DataStories-UniPi/VCRA.

In order to discover encountering vessels, the data points should be aligned at the temporal axis. However, as already mentioned, this is not the rule in AIS. To overcome this problem, the location of each vessel at a specific time is extrapolated from its two previous positions. This method has low latency but may result in false-positive results as the distance from the last known position grows. If no location is reported between two consecutive time-slices, the object is not further considered in the vessel encountering process.

Our data flow consists of three Kafka topics for temporarily storing the raw AIS data stream, the pairs of encountering vessels, and their corresponding locations and CRI, respectively.

3 SYSTEM DEMONSTRATION

In this section, we present the demo specifications of *VesselVision* via real-world data-streams. For the demonstration of our system, we use three real-world datasets, referred to as the Brest [7], Piraeus [11], and MarineTraffic[2] dataset, respectively, which consist of vessel positions (and other related information) collected through AIS within Brest Bay in France, port of Piraeus and nearby area in Greece, and the Aegean sea in Greece, respectively.

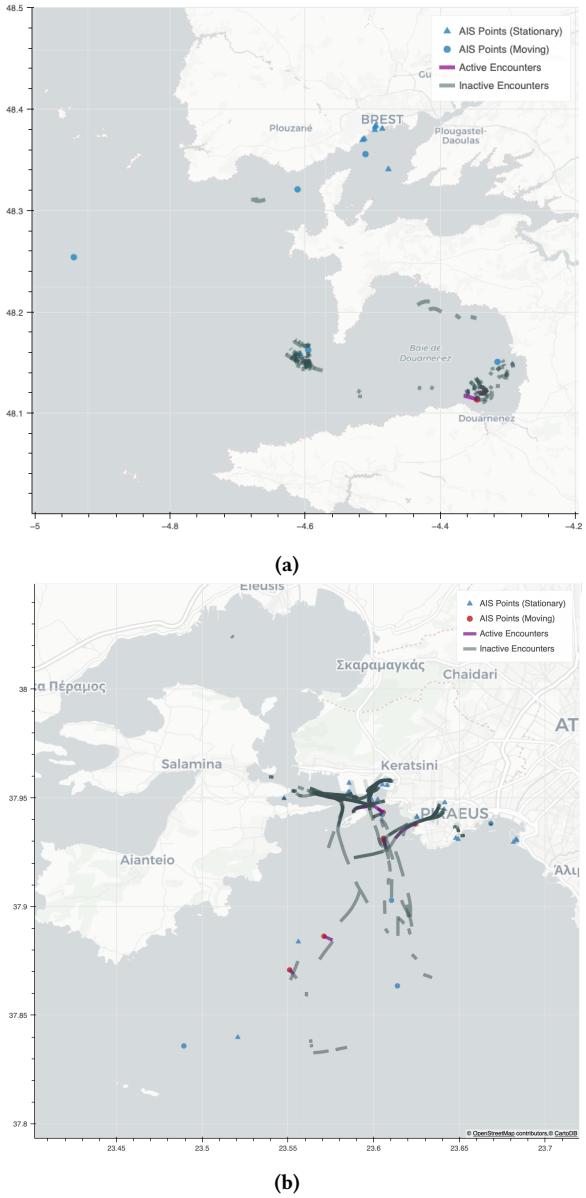
For the temporal alignment we use a buffer of 60 sec. for the Brest and Piraeus datasets, while for the VE module, we set $d_{max} = 1$ n.m. and $t_{min} = 60$ sec with an alignment rate equal to 30 sec. In the case of the MarineTraffic dataset, due to the sparsity of the vessels' trajectories we use a buffer of 360 sec. and set $t_{min} = 360$ sec., with an alignment rate equal to 180 sec.

The core concept of our demonstration is two-fold: on the one hand, the user has the ability to explore encountering vessels and assess their CRI, while on the other hand, to discover areas of interest that are associated with high CRI. Following this approach, we formulate a working hypothesis, according to which vessel encounters may correspond to known seaways/anchorages [10], indicating that further investigation is required.

The functionality of our system is divided into three panels: "Live Feed", "Analytics (Spatial Grid)", and "Analytics (Vessels' Trajectories)". The first panel consumes the data stream and visualizes the encountering vessels with their associated CRI (c.f., Fig. 3a and 3b), while the second and third panels provide a historical view on high-risk (in terms of CRI) areas (c.f., Fig. 4) and vessel trajectories, respectively. To provide additional degrees of freedom, all panels include controls for the CRI alert threshold and spatial resolution. Additionally, the latter two panels offer an extra control for adjusting the temporal horizon of the stored streaming AIS data, enabling more comprehensive and detailed analytics.

In particular, Figure 3 illustrates the encountering processes on the Brest and Piraeus datasets, respectively. We observe that the encounterings of the Brest dataset indeed tend to be formed near ports and known anchorages [10], whereas the encounterings of the Piraeus dataset concentrate more on intra-port traffic, particularly on the ports' entrance/exit spots, with less emphasis on anchorages.

At a higher level of detail, Figure 4 illustrates high-risk (in terms of CRI; "Analytics (Spatial Grid)" mode) areas of interest in Brest Dataset. While it leads us to similar conclusions, it offers quick insight on high-risk regions of the whole spatial region of the



(a)

(b)

Figure 3: Visualisation of encountering vessels with their corresponding CRI in (a) Brest; and (b) Piraeus datasets.

dataset, and for the sake of completeness one can adjust the zoom level for a more detailed view.

Deploying computationally demanding applications poses significant challenges due to the sheer scale and complexity involved. To ensure great performance and scalability, *VesselVision* provides a viable and practical solution to these issues. Figure 5 illustrates such a scenario on the MarineTraffic dataset, where we monitor vessel encounterings at the Aegean sea, Greece. For the purpose of our demonstration, the VesselVision user will have the opportunity to realize that even for such a large dataset, the latency of our tool appears to be consistently lower than the average sampling rate

[2]The dataset was kindly provided by MarineTraffic

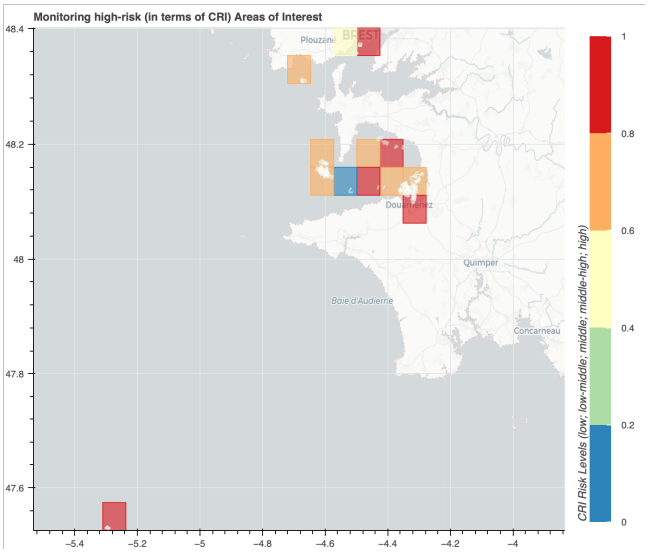


Figure 4: Visualisation of high-risk (in terms of CRI) areas of interest in Brest Dataset.

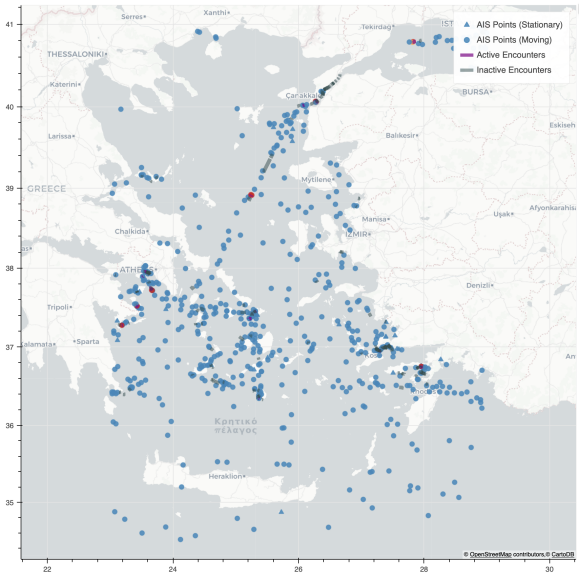


Figure 5: Deploying VesselVision on computationally demanding scenarios - an example using the MarineTraffic dataset.

of the experimental datasets , showing that Vesselvision can easily handle real-world AIS streams.

More details about VesselVision (source code, scenario demonstrations, etc.) are available at the dedicated webpage: https://www.datastories.org/vesselvision.

4 CONCLUSIONS

In conclusion, in this paper we demonstrated VesselVision, an MDA framework focusing on discovering encountering vessels and associating them with an index corresponding to their collision risk. As it was shown through real-world AIS streams, it can serve as a useful tool for maritime safety in real-time environment. In our future plans, we aim to extend our system by incorporating the notion of location prediction, so as to predict vessels' CRI in short-term, thus shifting to the so-called Vessel Collision Risk Forecasting (VCRF) problem [12]. Additionally, we aim to extend VesselVision by integrating state-of-the-art methods in collision avoidance [4] and visualizing suggested alternative itineraries. Last but not least, our system could be tuned to work in other than maritime use cases, including urban and aviation traffic, taking into account the special characteristics of those domains [3, 13].

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